

Lecture 6

Combining Systems: Entanglement

Art: *This is a pretty friendly place after all. Except for Minus One, I don't see too many loners*.

Lenny: *Mingling is only natural at a place like this. And not just because it's cramped. Just keep track of your wallet and don't get too entangled*.

6.1 MATHEMATICAL INTERLUDE: TENSOR PRODUCTS

6.1.1 Meet Alice and Bob

Figuring out how systems combine to make bigger systems is a large part of what we do in physics. I hardly need to tell you that an atom is a collection of nucleons and electrons, each of which could be considered a quantum system in its own right.

When talking about composite systems, it's easy to get bogged down in formal language like *System A* and *System B.* Most physicists prefer lighter-weight, informal language instead, and *Alice* and *Bob* have become near-universal substitutes for *A* and *B*. We can think of Alice and Bob as purveyors of composite systems and laboratory setups of every description. Their inventory and expertise are limited only by our imaginations, and they